

Youyun Zheng

Home Address:
260 Aspinwall Avenue
02445, Brookline, MA, USA
1-404-747-2283
youyunzheng@hms.harvard.edu

PROFESSIONAL PROFILES

Google Scholar: [Youyun Zheng](#)

LinkedIn: <https://www.linkedin.com/in/youyun-peter-zheng-970777106/>

EDUCATION

Emory University

BS in Biology, BA in Computer Science

Aug. 2014 – May. 2018

GPA 3.91/4

Harvard Medical School

MD/PhD Program (Pathways track, Bioinformatics and Integrative Genomics PhD Program)

Aug. 2020 – Present

RESEARCH EXPERIENCE

Dana Farber Cancer Institute

Graduate Student

*Department of Cancer Biology and Medical Oncology
(Principal Investigator: Dr. Rameen Beroukhim)*

Dec. 2022 - Present

- Leveraging large cohorts of cancer whole genome sequencing data to study the mechanism of insertions in structural variant breakpoints
- Investigated the tumor microenvironment of pediatric low grade brain tumors using single-cell spatially-resolved transcriptomics
- Analyzed a longitudinal multi-omic dataset from a glioblastoma clinical trial to characterize tumor heterogeneity, evolution and spatial architecture
- Mentored junior lab members on projects focused on demonstrating and rectifying ancestry-related bias in genomic data analysis

Boston Children's Hospital

Rotation Student

*Department of Pediatric Oncology
(Principal Investigator: Dr. Vijay Sankaran)*

Sept. 2022 –Nov. 2022

- Leveraged findings from previous large scale population genetics studies to identify cell types of interest for over 40 diseases in single cell RNA sequencing data sets
- Using machine learning models for cell type prediction in various state of the art large scale single cell tissue atlases

Dana Farber Cancer Institute

Rotation Student

*Department of Medical
Oncology*

Youyun Zheng

(Principal Investigator: Dr. Eliezer Van Allen)

Jun. 2020 – May 2021

- Using LSTM models to predict chronological sequence of occurrence for single nucleotide variations in a cohort of 1k melanoma whole exome sequencing data
- Identify alternative approaches to formatting training data to reduce sparsity for better performance and interpretability of prediction results

Massachusetts General Hospital

Sept. 2020 – Jun. 2022

Student Researcher

MESH Incubator

(Principal Investigator: Dr. Marc Succi)

- Leveraging natural language processing to predict radiology utilization based on ER nurse triage notes for operation efficiency
- Investigate differential imaging utilization rates and their socioeconomic impact in patients with and without limited English proficiency

Memorial Sloan Kettering Cancer Center

Jun. 2018– Jun. 2020

Bioinformatics Technician

Center for Molecular Oncology

(Principal Investigator: Dr. Michael Berger)

- Leveraged cerebrospinal fluid (CSF) cell-free DNA (cfDNA) sequencing to capture the real-time mutational landscape of gliomas, thereby providing a minimally invasive genomic testing alternative that allows for longitudinal disease monitoring and treatment guidance.
- Analyzed mutation data to confirm that our rectal cancer organoids reflect not only the mutational spectrum of their paired derivation cancer tissue but also a cohort of 291 rectal cancer specimens collected at Memorial Hospital.
- Led the effort in identifying genomic correlates of response in a phase II clinical trial on the usage of pembrolizumab in patients with advanced adrenocortical carcinoma.
- Contributed to the development and calibration of a genomics based machine learning tumor type classifier and implemented the classifier on the web sign-out portal of the MSKCC pathology department.
- Currently leading the analysis on two similar projects using ultra-deep sequencing plasma cfDNA liquid biopsy to track disease progression and identify treatment resistance mechanisms in ongoing clinical trials.
- Led the development of the rearrangement and fusion calling pipelines for our prospective ultra-deep plasma cfDNA and targeted FFPE RNA sequencing assay

MD Anderson Cancer Center

Jun. 2017 – Aug. 2017

Summer Research Student

Department of Genomic Medicine

(Principal Investigator: Dr. Andrew Futreal)

- Categorized genomic subtypes of osteosarcoma leveraging machine learning algorithms and whole genome and transcriptome sequencing data
- Conducted artifact filtering and consensus calling utilizing rearrangement (Breakdancer, Lumpy, BRASS) and fusion (MapSplice, Tophat-Fusion, FusionCatcher) calling algorithms
- Performed detailed analysis on structural variation profiles of osteosarcoma samples

Youyun Zheng

Emory University Rollins School of Public Health

Aug. 2016 – May. 2018

Undergraduate Researcher

Department of Biostatistics and Bioinformatics

(Principal Investigator: Dr. Zhengjia Chen)

- Examined and calculated key operating characteristics of Phase I cancer clinical trial with standard 3+3 and EWOC-NETS design
- Implemented back end code using Markov Chain Monte Carlo and other probability calculations
- Built front end of two software for displaying simulation and calculation results using Rshiny
- Achieved sufficiency in R language and various packages through self-learning utilizing various online resources

MD Anderson Cancer Center

Jun. 2016 – Aug. 2016

Summer Research Student

Department of Bioinformatics and Computational Biology

(Principal Investigator: Dr. Arvind Rao)

- Identified associations between MRI imaging features and protein expression levels in breast cancer for imaging guided diagnostics
- Expedited research projects by automating chart review of over 2,000 MRI scans and other related patient information
- Performed high-dimensional regression and multiple testing correction analysis

Emory University School of Medicine

Jan. 2015 – May. 2016

Undergraduate Researcher

Department of Pharmacology

(Principal Investigator: Dr. Haian Fu)

- Validated and gained insights into oncologic protein-protein interaction (PPI) between phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha (PIK3CA) hotspot mutant E545K and insulin receptor substrate 1 (IRS1) as target for personalized therapies
- Interpreted and illustrated experimental results through a 15-page progress report and poster presentation

HONORS AND AWARDS

**MD Anderson Cancer Center CPRIT-CURE Summer
Research Scholar**

Jun. 2017 – Aug. 2017

**MD Anderson Cancer Center Summer Undergraduate
Research Program (SURP)**

Jun. 2016 – Aug. 2016

Emory Civic Scholar Program

May. 2016 – May. 2017

Phi Beta Kappa Honor Society

Sept. 2017

PUBLICATIONS

Youyun Zheng

- Rosen, E. Y., Won, H. H., **Zheng, Y.**, Cocco, E., Selcuklu, D., Gong, Y., ... & Drilon, A. The evolution of RET inhibitor resistance in RET-driven lung and thyroid cancers. *Nat Commun* 13, 1450 (2022). <https://doi.org/10.1038/s41467-022-28848-x>
- Raj, N., **Zheng, Y.**, Hauser, H., Chou, J., Rafailov, J., Bou-Ayache, J., Sawan, P., Chaft, J., Chan, J., Perez, K., Rudin, C., Tang, L., & Reidy-Lagunes, D. (2021). Ribociclib and everolimus in well-differentiated foregut neuroendocrine tumors, *Endocrine-Related Cancer*, 28(4), 237-246. <https://doi.org/10.1530/ERC-20-0446>
- Wu, C.-C., Beird, H.C., Andrew Livingston, J., Advani, S., Mitra, A., Cao, S., Reuben, A., Ingram, D., Wang, W.-L., Ju, Z., Hong Leung, C., Lin, H., **Zheng, Y.**, Roszik, J., Wang, W., Patel, S., Benjamin, R.S., Somaiah, N., Conley, A.P., Mills, G.B., Hwu, P., Gorlick, R., Lazar, A., Daw, N.C., Lewis, V., Futreal, P.A., 2020. Immuno-genomic landscape of osteosarcoma. *Nat. Commun.* 11, 1008. <https://doi.org/10.1038/s41467-020-14646-w>
- Penson, A., Camacho, N., **Zheng, Y.**, Varghese, A.M., Al-Ahmadie, H., Razavi, P., Chandarlapaty, S., Vallejo, C.E., Vakiani, E., Gilewski, T., Rosenberg, J.E., Shady, M., Tsui, D.W.Y., Reales, D.N., Abeshouse, A., Syed, A., Zehir, A., Schultz, N., Ladanyi, M., Solit, D.B., Klimstra, D.S., Hyman, D.M., Taylor, B.S., Berger, M.F., 2019. Development of Genome-Derived Tumor Type Prediction to Inform Clinical Cancer Care. *JAMA Oncology*. <https://doi.org/10.1001/jamaoncol.2019.3985>
- Raj, N., **Zheng, Y.**, Kelly, V., Katz, S.S., Chou, J., Do, R.K.G., Capanu, M., Zamarin, D., Saltz, L.B., Ariyan, C.E., Untch, B.R., O'Reilly, E.M., Gopalan, A., Berger, M.F., Olino, K., Segal, N.H., Reidy-Lagunes, D.L., 2019. PD-1 Blockade in Advanced Adrenocortical Carcinoma. *J. Clin. Oncol.* JCO.19.01586. <https://doi.org/10.1200/JCO.19.01586>
- Ganesh, K., Wu, C., O'Rourke, K.P., Szeglin, B.C., **Zheng, Y.**, Sauv  , C.-E.G., Adileh, M., Wasserman, I., Marco, M.R., Kim, A.S., Shady, M., Sanchez-Vega, F., Karthaus, W.R., Won, H.H., Choi, S.-H., Pelossof, R., Barlas, A., Ntiamoah, P., Pappou, E., Elghouayel, A., Strong, J.S., Chen, C.-T., Harris, J.W., Weiser, M.R., Nash, G.M., Guillem, J.G., Wei, I.H., Kolesnick, R.N., Veeraraghavan, H., Ortiz, E.J., Petkovska, I., Cercek, A., Manova-Todorova, K.O., Saltz, L.B., Lavery, J.A., DeMatteo, R.P., Massagu  , J., Paty, P.B., Yaeger, R., Chen, X., Patil, S., Clevers, H., Berger, M.F., Lowe, S.W., Shia, J., Romesser, P.B., Dow, L.E., Garcia-Aguilar, J., Sawyers, C.L., Smith, J.J., 2019. A rectal cancer organoid platform to study individual responses to chemoradiation. *Nat. Med.* 25, 1607–1614. <https://doi.org/10.1038/s41591-019-0584-2>
- Miller, A.M., Shah, R.H., Pentsova, E.I., Pourmaleki, M., Briggs, S., Distefano, N., **Zheng, Y.**, Skakodub, A., Mehta, S.A., Campos, C., Hsieh, W.-Y., Selcuklu, S.D., Ling, L., Meng, F., Jing, X., Samoila, A., Bale, T.A., Tsui, D.W.Y., Grommes, C., Viale, A., Souweidane, M.M., Tabar, V., Brennan, C.W., Reiner, A.S., Rosenblum, M., Panageas, K.S., DeAngelis, L.M., Young, R.J., Berger, M.F., Mellinghoff, I.K., 2019. Tracking tumour evolution in glioma through liquid biopsies of cerebrospinal fluid. *Nature* 565, 654–658. <https://doi.org/10.1038/s41586-019-0882-3>

Youyun Zheng

- Chen, Z.*, **Zheng, Y.***, Wang, Z., Kutner, M., Curran, W.J., Kowalski, J., 2018. Interactive calculator for operating characteristics of phase I cancer clinical trials using standard 3+3 designs. *Contemp. Clin. Trials Commun.* 12, 145–153.
<https://doi.org/10.1016/j.conctc.2018.10.006> (Co-first author)
- Lehrer, M., Bhadra, A., Aithala, S., Ravikumar, V., **Zheng, Y.**, Dogan, B., Bonaccio, E., Burnside, E.S., Morris, E., Sutton, E., Whitman, G.J., Net, J., Brandt, K., Ganott, M., Zuley, M., Rao, A., 2018. High-dimensional regression analysis links magnetic resonance imaging features and protein expression and signaling pathway alterations in breast invasive carcinoma. *Oncoscience* 5, 39–48. <https://doi.org/10.18632/oncoscience.397>

TEACHING AND MENTORSHIP

Class Mentor and TA for Emory Emergency Medical Service training course

May. 2016 – Aug. 2017

Rotation Student (Ola Oni) Tri-I CBM PhD student

Jul. 2019 – Nov. 2019

COMMUNICATIONS AND PRESENTATIONS

POSTER COMMUNICATION

Clinical validation of a genomics-based classifier to predict tissue of origin from targeted tumor sequencing.

*American Association for Cancer Research Annual Conference
Atlanta, GA, USA - 04/2019*

Identifying Patterns in Rearrangement Profiles in Osteosarcoma Patients.

*MD Anderson Cancer Center Summer Undergraduate Research Symposium
Houston, TX, USA - 08/2017*

Calculation of 5 Key Statistical Properties for Phase I Cancer Clinical Trial Designs.

*Emory University Undergraduate Research Symposium
Atlanta, GA, USA - 05/2017*

MR Images and Proteomics Data Predicting Diseases, Functions and Canonical Pathways.

*MD Anderson Cancer Center Summer Undergraduate Research Symposium
Houston, TX, USA - 08/2016*

Screening of Anti-cancer Drug Based on the Tumor Specific Interaction Between p110 α -E545K and IRS1.

*Emory University Undergraduate Research Symposium
Atlanta, GA, USA - 05/2016*

ORAL PRESENTATION

Interactive Software for Dose Calculation and Simulation of Phase I Cancer Clinical Trial Using EWOC-NETS Design.

Youyun Zheng

*Emory University Department of Biology Honors Thesis Defense
Atlanta, GA, USA - 05/2018*

CONFERENCE ABSTRACTS

Association of Language with Image Utilization in an Emergency Department.

Radiological Society of North America (1st author)

Chicago, IL, USA - 11/2021

Next-level Imaging Triage: Predicting Imaging Utilization based on Emergency Department Initial Triage Notes using Machine Learning and Natural Language Processing.

American Society Emergency Radiology (1st author)

Tempa, FL, USA - 09/2021

Say it on Rounds: Association of Limited English Proficiency (LEP) with Increased Imaging Utilization in an Emergency Department.

American Society Emergency Radiology (2nd author)

Tempa, FL, USA - 09/2021

Association of Language with Length of Stay in an Emergency Department.

American Society Emergency Radiology (1st author)

Tempa, FL, USA - 09/2021

Efficacy and safety of pembrolizumab in patients with advanced adrenocortical carcinoma.

American Society of Clinical Oncology Annual Conference (2nd author)

Baltimore, MD, USA - 05/2019

Clinical validation of a genomics-based classifier to predict tissue of origin from targeted tumor sequencing.

American Association for Cancer Research Annual Conference (1st author)

Atlanta, GA, USA - 04/2019

A bioinformatics framework for high-sensitivity detection and monitoring of oncogenic gene fusions in plasma cfDNA.

American Association for Cancer Research Annual Conference (7th author)

Atlanta, GA, USA - 04/2019

Parallel genomic and immune profiling of relapsed and metastatic osteosarcoma to reveal bases of low immunogenicity.

American Society of Clinical Oncology Annual Conference (12th author)

Chicago, IL, USA - 05/2018

Genome and transcriptome profiling of relapsed and metastatic osteosarcoma.

American Society of Clinical Oncology Annual Conference (12th author)

Chicago, IL, USA - 05/2018

TECHNICAL SKILLS

- **Programming:** R, Bash, Python, Java, MySQL and PHP (*ordered by expertise*).
- **Analysis:** variant calling (SNVs, SCNA, SVs), MSI calling, RNAseq, capture RNAseq, enrichment analysis, sequencing analysis.
- **Software:** Vardict, FACETS, Manta, Delly, MSISensor, LOHHLA, Arriba, FusionCatcher, STAR, samtools, IGV, R studio, GitHub, VSCode, Microsoft Works, illustrator, GIMP
- **Operative systems:** Linux, OS X, Windows.

CERTIFICATIONS AND COURSES

Data mining

*Emory University, Department of Computer Science
Atlanta, USA - 01/2018.*

Data structures and algorithms

*Emory University, Department of Computer Science
Atlanta, USA - 01/2015.*

Database systems

*Emory University, Department of Computer Science
Atlanta, USA - 01/2016.*

Genetics: a human perspective

*Emory University, Department of Biology
Atlanta, USA - 01/2018.*

Advanced Emergency Medical Technician (AEMT)

*Emory Emergency Medical Service, Emory University Division of Public Safety
Atlanta, USA - 08/2017.*

SERVICE AND LEADERSHIP EXPERIENCES

Emory Emergency Medical Service

Aug. 2015 – May. 2018

*Supervisor Advanced EMT, Field Training Officer
Emory University Division of Public Safety*

- Attained Georgia and national license after completing one year of medical training
- Managed patient care as an Advanced Emergency Medical Technician in prehospital setting
- Mentored the incoming EMT class acting as instructor in lab sessions to facilitate teaching
- Trained and evaluated new members for quality assurance and improvements in patient care
- Supervised primary unit in 24 hour shifts and provided administrative and medical support

Center for Black Women Wellness

Medical Volunteer

Youyun Zheng

Safety Net Clinic

Sep. 2016 – Dec. 2017

- Facilitated patient care in free clinic by taking vitals and medical history of incoming patients
- Developed patient rapport through communication and lessened workload of physicians

Winship Cancer Institute

Sep. 2016 – May. 2017

Medical Volunteer

Infusion Pharmacy

- Accelerated inventory process by restocking equipment for making chemo-treatments
- Simplified lab technicians' working processes in a sterile working environment